Name: _____

Question	Points	Score
1	6	
2	7	
3	5	
4	6	
5	7	
6	6	
7	8	
Total:	45	

1. Write your name on the front page of your exam.
2. Read each question carefully, and answer each question completely.
3. Write your solutions clearly in the exam sheet.
4. Show all of your work; no credit will be given for unsupported answers.
5. You may use the result of one part of the problem to prove a later part, even if you have not completed the earlier part.
6. You may use major theorems proved in class, but not when the whole point of the problem is to reproduce the proof of a theorem covered in class or in the textbook. When in doubt, ask us whether you are allowed to use a particular result.

1. Suppose G is a group of order $5 \cdot 7 \cdot 11$.

   (a) (4 points) Prove that $G \simeq (\mathbb{Z}/11\mathbb{Z}) \rtimes (\mathbb{Z}/35\mathbb{Z})$.

   (b) (2 points) Prove that there are exactly two such groups up to an isomorphism.

2. Suppose G is a finite group and p is a prime divisor of $|G|$. Let $\mathrm{Syl}_p(G)$ denote the set of all Sylow p-subgroups of G, and

$$O_p(G) := \bigcap_{P \in \mathrm{Syl}_p(G)} P.$$

(a) (3 points) Prove that $O_p(G)$ is the largest normal p-subgroup of G; that means $O_p(G)$ is a normal p-subgroup and if N is a normal p-subgroup of G, then $N \subseteq O_p(G)$.

(b) (1 point) Suppose H is a subgroup of G. Prove that there exists a function

$$f \colon \mathrm{Syl}_p(H) \to \mathrm{Syl}_p(G)$$

such that $f(P) \cap H = P$.

(c) (2 points) Prove that if $|\mathrm{Syl}_p(H)| = |\mathrm{Syl}_p(G)|$, then

$$O_p(H) = O_p(G) \cap H.$$

(**Hint**. Show that the function f given in the previous part is a bijection)

(d) (1 point) Prove that if $O_p(G) = 1$ and $\overline{P}$ is a non-trivial p-subgroup of G, then

$$|\mathrm{Syl}_p(N_G(\overline{P}))| < |\mathrm{Syl}_p(G)|.$$

3. (5 points) Find the number of conjugacy classes of $\mathrm{GL}_3(\mathbb{F}_q)$, where q is a power of a prime number.

4. Suppose A is a unital commutative ring and $\mathrm{Spec}(A)$ is the set of all the prime ideals of A. For an A-module M and $\mathfrak{p} \in \mathrm{Spec}(A)$, let $M_{\mathfrak{p}}$ be the localization of M at $\mathfrak{p}$. Let

$$\mathrm{supp}\, M := \{\mathfrak{p} \in \mathrm{Spec}(A) \mid M_{\mathfrak{p}} \neq 0\}.$$

(a) (3 points) Prove that for a finitely generated A-module M,

$$\mathrm{supp}\, M = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathrm{ann}(M) \subseteq \mathfrak{p}\},$$

where $\mathrm{ann}(M)$ is the annihilator of M.

(b) (3 points) For a positive integer d, let

$$U_d(M) := \{\mathfrak{p} \in \mathrm{Spec}(A) \mid M_{\mathfrak{p}} \text{ can be generated by } d \text{ elements}\}.$$

Prove that if M is finitely generated and $\mathfrak{p}_0 \in U_d(M)$, then there exists $a_0 \in A \setminus \mathfrak{p}_0$ such that

$$\{\mathfrak{q} \in \mathrm{Spec}(A) \mid a_0 \notin \mathfrak{q}\} \subseteq U_d(M).$$

(**Hint**. Suppose $N_{\mathfrak{p}_0} = M_{\mathfrak{p}_0}$ where $N = \langle x_1, \ldots, x_d \rangle \subseteq M$. Consider $\mathrm{supp}(M/N)$.)

5. Suppose D is a Noetherian integral domain, F is a field of fractions of D, $\mathfrak{a} \trianglelefteq D$ is an ideal of D, and M is a finitely generated D-submodule of F. Suppose $\mathfrak{a}M = D$, where

$$\mathfrak{a}M = \{\sum_{i=1}^{n} c_i x_i \mid c_i \in \mathfrak{a}, x_i \in M\}.$$

(a) (4 points) Prove that M is projective.

(b) (3 points) Prove that $\mathfrak{a} \otimes_D M \simeq D$.

6. Let $f(x) = x^4 + x + 1 \in \mathbb{F}_2[x]$. Let $E = \mathbb{F}_2[x]/\langle f \rangle$.

(a) (4 points) Show that E is a field.

(b) (2 points) Let $\alpha = (x^2 + x) + \langle f \rangle \in E$. Find the number of elements in $\mathbb{F}_2(\alpha)$.

7. (8 points) Suppose that K/F is a Galois extension of fields with

$$\mathrm{Gal}(K/F) \cong S_n,$$

for some $n \geq 2$. Show that there is an irreducible polynomial $f(x) \in F[x]$ with $\deg f(x) = n$, such that K is the splitting field of $f(x)$ over F.

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Good Luck!